

AROGYA KOWSHIK KUPPALA

SOFTWARE ENGINEER

London, United Kingdom | +44 779 937 3241 | Email: arogyakowshik6@gmail.com | LinkedIn: [linkedin.com/in/arogya-kowshik-5732ab387](https://www.linkedin.com/in/arogya-kowshik-5732ab387) | GitHub: <https://github.com/arogyakowshik6>

PROFILE SUMMARY

Team player and results-driven Frontend Developer with an MBA in Global Business and a BEng in Electronics and Communication Engineering. Experienced in building scalable React.js, Next.js, TypeScript, and JavaScript applications, with expertise in Redux, REST API integration, SSR, performance optimisation, and automated testing. Proven track record delivering production-grade solutions across banking, e-commerce, and healthcare environments using Agile methodologies and modern frontend development practices.

KEY SKILLS

Frontend Development

- Single Page Applications (SPA), Server-Side Rendering (SSR), Incremental Static Regeneration (ISR)
- React.js (v18+), Next.js (App Router), Vue.js, TypeScript, JavaScript (ES6+)
- HTML5, CSS3, Responsive Web Design, Cross-browser Compatibility
- Web Accessibility Standards and Component-Based Architecture

Backend & Database Development

- Database Design, Query Optimisation and Data Management
- RESTful API Integration, Data Handling and Backend Services
- Authentication, Authorisation and JWT Implementation
- Python, PHP and SQL Development
- MySQL, PostgreSQL and MongoDB

Cloud, State Management & Performance

- Core Web Vitals, Rendering Optimisation and Responsive Interfaces
- Performance Optimisation, Lazy Loading and Code Splitting
- Redux Toolkit, Context API, Zustand
- React Query, SWR, Axios
- AWS and Microsoft Azure

Testing, Tools & Agile Delivery

- Design Systems, Tailwind CSS, Material UI (MUI), Bootstrap and Styled Components
- Debugging, Unit Testing, End-to-End Testing and CI/CD Practices
- Agile Scrum Methodologies, Sprint Planning and Code Reviews
- Jest, React Testing Library, Cypress, Playwright and Vitest
- Git, GitHub, Jira and Confluence

EDUCATION AND CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services (AWS) | May 2026

Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft | May 2026

MBA- Global Business | Coventry University | May 2026

Bachelor of Technology in Electronics and Communication Engineering | Vignan University | August 2018

TECHNICAL PROJECTS

Mashreq Bank-Digital Banking Platform

Technologies: React.js, Next.js, JavaScript, Redux, REST APIs, HTML5, CSS3, Git, Jira

- Developed and maintained a scalable digital banking platform for Mashreq Bank, delivering responsive web and mobile-first interfaces using React.js and Next.js within a component-based architecture.
- Built reusable, modular UI components from Figma designs, ensuring pixel-perfect implementation, accessibility compliance (WCAG standards), and consistent design system adherence.
- Integrated RESTful APIs for core banking functionality including account management, payments, and transaction analytics, utilising Redux for predictable global state management and data flow control.
- Implemented Server-Side Rendering (SSR) and Static Site Generation (SSG) with Next.js to improve SEO performance, reduce Time to First Byte (TTFB), and enhance Core Web Vitals by approximately 30%.
- Participated in Agile/Scrum ceremonies including sprint planning, stand-ups, and retrospectives, while contributing to unit testing, code reviews, and technical documentation in Confluence to ensure maintainable, production-grade code.

Liverpool-B2B E-Commerce Marketplace (500,000+ Products)

Technologies: React.js, Redux Toolkit, REST APIs, Axios, HTML5, CSS3, Bootstrap, JavaScript

- Developed and maintained a high-performance Single Page Application (SPA) using React.js and Redux, supporting product discovery, order management and supplier workflows for 10,000+ business users.
- Optimised frontend performance through lazy loading, component memoisation, image optimisation and reduced unnecessary re-renders, decreasing page load times by 20%.
- Implemented efficient state management and asynchronous data handling using Redux and Axios, improving application scalability and responsiveness.
- Integrated RESTful APIs to deliver real-time product catalogue data, order tracking functionality and supplier dashboard features across the platform.
- Built responsive, mobile-friendly user interfaces using Bootstrap, HTML5 and CSS3, ensuring cross-browser compatibility and accessibility standards.
- Applied form validation, error handling and debugging techniques to improve application stability and user experience.
- Contributed to Agile ceremonies, peer code reviews, unit testing and defect resolution to maintain high-quality software delivery.

Manipal Hospital Portal-Patient Engagement Platform

Technologies: React.js, Redux, RxJS, REST APIs, Jest, React Testing Library (RTL), JavaScript

- Developed modular and reusable React components for appointment scheduling, electronic medical record (EMR) access, patient profile management and secure messaging functionality.
- Implemented scalable state management using Redux and RxJS to support asynchronous data streams, real-time notifications and patient alert systems.
- Integrated RESTful APIs to enable secure access to patient information, appointment data and healthcare communication services.
- Applied secure coding principles, JWT authentication and HTTPS protocols to protect sensitive healthcare data and support compliance requirements.
- Enhanced application performance through lazy loading, component optimisation and responsive design techniques across desktop and mobile platforms.
- Created and executed unit and component tests using Jest and React Testing Library, improving application reliability and reducing production defects.
- Collaborated with cross-functional teams within an Agile environment to deliver new features, resolve defects and continuously improve user experience.

PROFESSIONAL EXPERIENCE

Software Engineer | VisionQuest Solutions | November 2020-January 2025

- Developed and maintained scalable frontend applications using React.js, Next.js, TypeScript, and JavaScript within Agile software development environments.
- Designed and implemented responsive, component-based user interfaces using HTML5, CSS3, Tailwind CSS, and Material UI, ensuring accessibility, usability, and cross-browser compatibility.
- Integrated RESTful APIs and third-party services to deliver secure, data-driven web applications across banking, e-commerce, and healthcare sectors.
- Managed application state using Redux Toolkit, Context API, and Zustand, while developing reusable components, custom hooks, and scalable frontend architectures to improve maintainability and accelerate feature delivery.
- Implemented Server-Side Rendering (SSR), Incremental Static Regeneration (ISR), and frontend performance optimisation techniques to enhance Core Web Vitals and overall user experience.
- Conducted unit, integration, and end-to-end testing using Jest, React Testing Library, Cypress, and Playwright, ensuring software quality, reliability, and production readiness.
- Participated in Agile ceremonies, code reviews, debugging, and technical documentation, collaborating with cross-functional teams to deliver high-quality software solutions.
- Delivered production-ready applications for enterprise clients within compliance-driven environments, supporting thousands of end users while adhering to development best practices.

Web Development Intern | Myneni's IT Solutions Pvt Ltd | December 2017-April 2018

- Developed a strong foundation in frontend development principles, coding standards and version control practices within a professional software development environment.
- Participated in testing, troubleshooting and quality assurance activities to improve application functionality and performance.
- Gained practical experience in responsive web design, web application architecture and software development lifecycles.
- Collaborated with development teams to understand project requirements and contribute to successful project delivery.
- Supported senior developers with frontend development tasks, bug fixing and user interface enhancements.
- Assisted in the development and maintenance of web-based applications using HTML, CSS and JavaScript.

References Available Upon Request